

*Calculators may be used in this examination but must not be used to store text.
Calculators with the ability to store text should have their memories deleted prior
to the start of the examination.*

UNIVERSITY OF BIRMINGHAM

School of Civil Engineering

Level C

**04 23632
(CE1SAM)**

Statics and Mechanics

Summer Examinations 2013

Time Allowed: 3 hours

Answer **FIVE** questions only

All questions carry equal marks. The allocation of marks within each question is stated in the right-hand margin.

1. For the structure shown in Figure Q1.

(a) Determine the support reactions [5]

(b) Show that the shear force F and bending moment M equations can be expressed as:

$$F = V_A - 12\langle x - 2 \rangle^0 - 2\langle x - 4 \rangle - \frac{1}{3}\langle x - 4 \rangle^2$$

$$M = V_A x - 12\langle x - 2 \rangle - \langle x - 4 \rangle^2 - \frac{1}{9}\langle x - 4 \rangle^3$$

Where the brackets $\langle \rangle$ are the Macaulay Brackets [10]

(c) Draw the shear force and bending moment diagrams of the beam AB. [5]

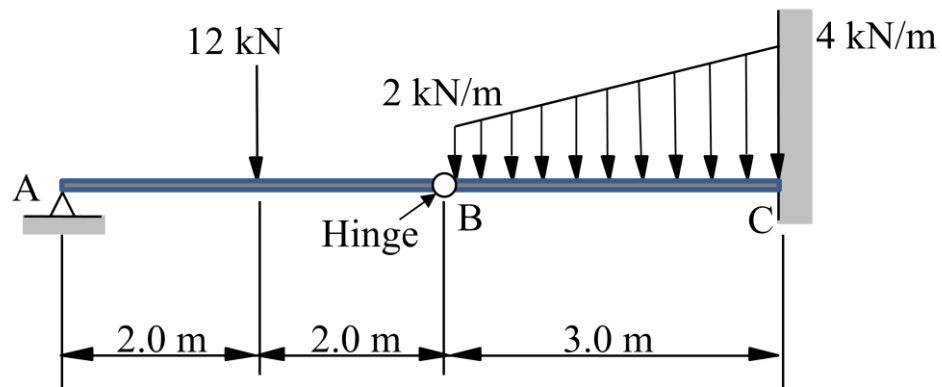


Figure Q1

2. (a) Three rods, positioned vertically, and each having a cross-sectional area of 600 mm^2 support a *rigid, weightless* bar, as shown in Figure Q2(a). Rod A is manufactured from aluminium ($E_{\text{aluminium}} = 60 \text{ kN/mm}^2$), rod B is manufactured from brass ($E_{\text{brass}} = 100 \text{ kN/mm}^2$) and rod C is manufactured from steel ($E_{\text{steel}} = 200 \text{ kN/mm}^2$).

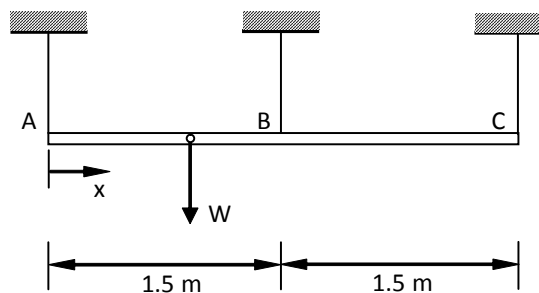


Figure Q2(a)

Determine the distance, x , from end A at which a weight, W , should be placed to keep the bar horizontal. [12]

- (b) The uniform cantilevers shown in Figure Q2(b) has a length L and a flexural rigidity EI .

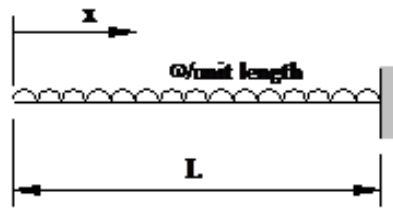


Figure Q2 (b)

Show that the deflection at the free end can be expressed as:

$$v_{x=0} = \frac{\omega L^4}{8EI}$$

[8]

3. Consider the shape illustrated in Figure Q3.

- (a) Show that the centroid of the shape is located at a distance of:

$$\bar{y} = \frac{27}{56}d = 0.482d$$

from the x-axis.

[8]

- (b) Hence, or otherwise show that the second moment of area of the shape about its centroid is given by:

$$I = 0.08 bd^3$$

[12]

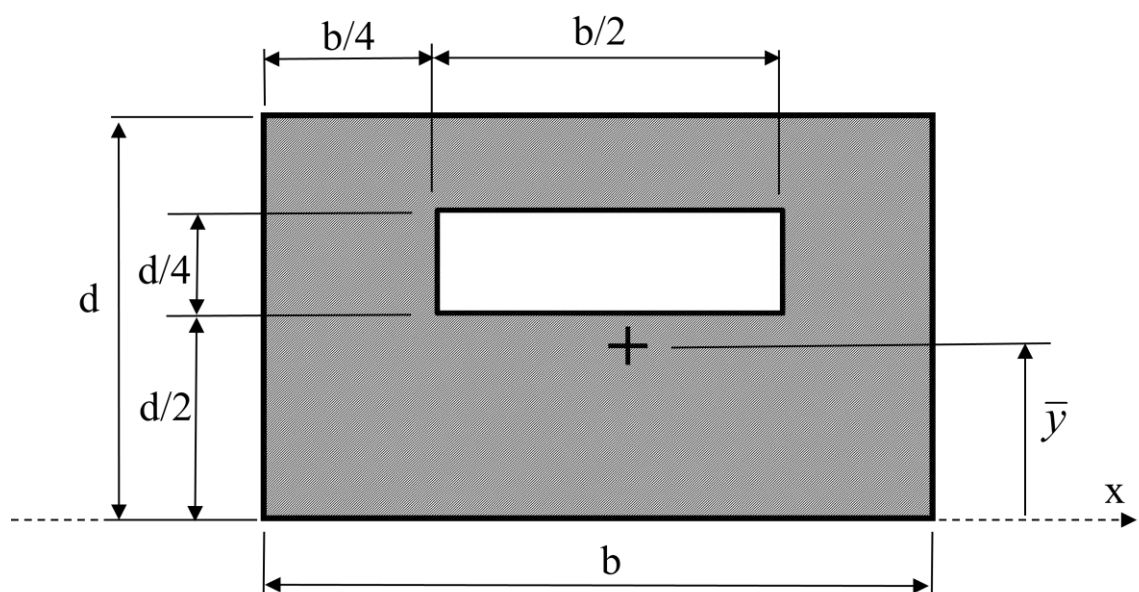


Figure Q3

4. (a) The doubly symmetrical I-section beam shown in Figure Q4 carries a shear force of 50kN. Show that the shear stress (τ) in the web is given by:

$$\tau = 30.11538 - 0.00041z^2 \text{ N/mm}^2$$

where z is the distance from the neutral axis.

[8]

- (b) What is the shear stress at the neutral axis and at the intersection of the web and flange?

[2]

- (c) Calculate the percentage of shear force carried by the web.

[10]

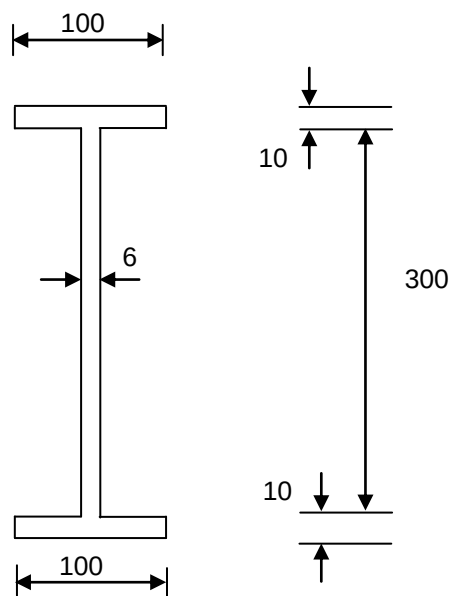


Figure Q4. (all dimensions in mm).